

This exam has 2 page(s), 11 questions, and 15 total marks. Answer all questions.

**Part 1: Fill in the Blanks**

*[2.5 marks — 1/2 mark each]*

1. (1/2 mark) \_\_\_\_\_ is unique because it spans across *all* other stages of the data engineering lifecycle.
  2. (1/2 mark) Unlike lifecycle stages, \_\_\_\_\_ are critical ideas relevant to every stage of the lifecycle.
  3. (1/2 mark) Under the Shared Responsibility Model, AWS is responsible for security \_\_\_\_\_ the cloud; the customer is responsible for security \_\_\_\_\_ the cloud.
  4. (1/2 mark) In early software engineering, data was often treated as a \_\_\_\_\_ rather than a primary asset.
  5. (1/2 mark) The modern era of data engineering was triggered by Google's publications on the \_\_\_\_\_ and MapReduce.
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**Part 2: Short Answer**

*[2.5 marks]*

1. (1 1/2 marks) What is the difference between a **functional** and a **non-functional** requirement? Give one example of each.

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2. (1 mark) What does it mean for a data engineer to “stand on the shoulders of giants”?

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**Part 3: Analytical Questions**

*[4 marks — 2 marks each]*

1. (2 marks) Analyze how the shift from **on-premises** systems to the **cloud** has changed the relationship between data architecture and data engineering.

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2. (2 marks) Explain why the data engineering lifecycle is considered **cyclical** rather than linear.

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**Part 4: Design Questions**

*[6 marks — 3 marks each]*

1. (3 marks) A data scientist reports that the marketing team needs **real-time** sales analysis but currently receives only a **daily batch** dump. Outline your **first three steps** using the “Thinking Like a Data Engineer” framework.

(a) **Step 1:**

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(b) **Step 2:**

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(c) **Step 3:**

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2. (3 marks) You are designing a data system for a **highly regulated industry** (e.g. healthcare). How does this affect your **Requirements Gathering** and the **Security** undercurrent?

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